Jay Sonani

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SUMMARY

Experienced **FDE**/AI Engineer with **6+ years** of experience and an active learning attitude, proficient in developing and shipping code/agents for improved performance, reliability, and scalability. Led full stack development, achieving a 25% faster loading time, streamlined information updates with cloud-based deployments and database management.

KEY SKILLSET

- **Python**
- TypeScript
- **REST API**
- GraphQL
- SDLC
- **Microservices**
- **AI Agents**, LLMs
- Prompt Engineering
- **Claude Code**, Skills
- MCP, Tool-calling
- **RAG** pipelines
- LangChain
- **AWS, GCP**
- Docker
- CI/CD
- Kubernetes
- **PostgreSQL**
- NodeJS
- **Conversational AI**
- Telephony, SIP
- ASR, TTS Engine
- **Voice Latency**
- **LLM** Integration
- SQL/NoSQL

PROFESSIONAL EXPERIENCE

Forward Deployed AI Engineer **Oct 2025 - Present**

PolyAI

- Developing **Voice AI Agents** for Enterprise customers using **Python**, LLMs, and AWS handling 500K+ calls/month at 70%+ containment rate, reducing escalations to human agents by 30%
- Leading end-to-end development and deployment of conversational AI systems for **10+ Enterprise clients** across hospitality, retail, healthcare, taking each from kickoff to go-live in 2-4 weeks, followed by Hypercare support
- Applying prompt engineering and **tool-calling** to maximize performance of proprietary **in-house LLMs**
- Configuring and tuning **ASR**(Nvidia) and **TTS**(ElevenLabs) pipelines, cutting word error rate by 15%
- Utilizing Agentic tools like Claude Code, CLAUDE.md, **SKILL.md**, Agent Skills, CursorRules, MCP Server, ADK
- Enhancing **real-time voice** interaction quality through Voice Activity Detection(VAD) and barge-in handling, reducing false interruptions by 40% and fewer mid-call hang-ups
- Integrating third-party APIs like OpenTable, Tripleseat for custom client asks: repeat caller detection, lead capture
- Writing custom user metrics and monitors on **Datadog**, and building analytical dashboards to monitor CSAT, NPS
- Reducing perceived end-to-end **latency** by 300ms through audio caching, guardrails, and runtime optimizations
- Performing **telephony** integration, migrations using **SIP** trunking on **Twilio** with Asterisk and Kamailio

AI Software Engineer **Apr 2024 - Feb 2025**

Replicant.ai

- Developed **Voice AI Agents** to automate Customer Service calls using **TypeScript** and **LLM** Components in a **fast-paced** multi-project **Agile** environment, making deployments in **GCP** buckets
- Orchestrated **RAG** pipeline and plugged **knowledge base** into the AI Agents to answer Customer FAQs
- Conducted 10+ daily peer **code reviews**, ensuring code quality and adherence to **best practices**
- Wrote unit, integration, and E2E tests using Jest for reliability, maintaining 95%+ code coverage.
- **Collaborated** daily with product managers, designers, QA and external stakeholders in a **cross-functional** team, maintaining technical documentation for code and system design solutions
- Debugged production issues with infrastructure monitoring like Datadog, deploying **microservices** on AWS
- Integrated customer **REST APIs** using OpenAPI specification and third-party CRMs(Salesforce, Zendesk) by pairing directly with customers' engineers, while using **PostgreSQL** relational database in back-end
- Participated in **on-call** rotation to provide support for issues and incidents 24x7 for **SaaS** product

Full Stack Developer **Apr 2023 - Apr 2024**

Pelmorex Corp - The Weather Network

- Orchestrated the development of 3 highly scalable, fast, reliable, real-time **web-based** applications
- Utilized Next.js, **React.js**, Node.js, Express, **TypeScript**, and MongoDB following **Agile** methodology
- Leveraged **AWS** (EC2, EBS, S3, Lambda, SNS, RDS) for cloud deployment, debugging production issues, optimizing performance and reducing deployment time by 40% for Single-Page Application(SPA)
- Designed efficient schema for **GraphQL** queries with Apollo in **Python** resulting in 14% faster page loads

- Leveraged **NodeJS** and Express for **back-end** development, achieving 40% faster response time for requests
- Participated in production support, including deployment, training and setting up and **incident** response
- Utilized GCP BigQuery for SQL, and deployed APIs on **CloudRun** as and when requested to ensure 100% uptime
- Collaborated with Product Managers and developers, using analytical and problem-solving skills, and making data-driven and detail-oriented decisions that improved project efficiency by 15%
- Implemented common **Design Patterns**, Object-oriented programming(OOP) concepts and SOLID principles

Software Developer Co-op **Sep 2022 - Dec 2022**

Tulip Retail

- Worked in an agile product development team to advance 3 **React** features for checkout and payment, integrated Tulip Software-as-a-Service (**SaaS**) platform with 4 retail brands, and demonstrated sprint features every 2 weeks
- Configured REST APIs with Python-based **AWS Lambda** functions to clean and categorise 100K+ news articles
- Performed **pair programming**, code review, and wrote unit tests using Jest achieving 90% code coverage
- Altered 2 APIs in Golang to use suitable data structures and algorithms resulting in a 13% performance boost
- Deployed the code on **multi-tenants** using Docker, Terraform and Kubernetes on Google Cloud
- Transformed, analyzed and broke down technical designs into tactical and executable blocks of work
- Participated in a company-wide Hackathon to develop a VS Code extension for assisting developers internally

Software Developer **Aug 2020 - Jul 2021**

Tata Consultancy Services

- Developed and maintained applications using **React**, Python and AWS, in a fast-paced work environment
- Initiated and executed a performance **optimization** initiative using **Redis caching**, achieving a 30% reduction in PostgreSQL database load and ensuring excellent application responsiveness in a challenging environment
- Utilized **AWS** (API Gateway, DynamoDB, SQS, IAM, Cognito) services eliminating development time by 35%
- Participated in 400+ peer **code reviews**, ensuring **high quality** code and adherence to **best practices**, followed Test-driven development
- Employed **Docker** and Jenkins for CI/CD pipelines, effectively reducing deployment times by 40%
- Maintained code on **Git** that included branching, pushing, merging, and conflict resolving while maintaining the Software Development Life Cycle for distributed highly performant web applications in collaboration with a team

EDUCATION

Master of Applied Computer Science **Sep 2021 - Dec 2022**

Dalhousie University

Bachelor of Computer Engineering **Aug 2016 - Aug 2020**

Gujarat Technological University

CERTIFICATIONS

NVIDIA-Certified Associate Generative AI LLMs *View* **Mar 2025**

Nvidia Gen AI LLMs

Generative AI Fundamentals *View* **Jan 2025**

Databricks

PROJECTS

Magical AI Agent *Code* **Apr 2025**

TypeScript, Vercel AI SDK, Playwright

Reverse Proxy Server *Code* **Apr 2025**

Golang, Redis, Caching

Inventory Management *Code* **Mar 2024**

TypeScript, Node.js, Jest

Gracious Givers *Code* **Aug 2022**

React, Node.js, Express.js, MongoDB

Eventify *Code* **Feb 2022**

AWS, CloudFormation, EC2, Lambda, S3, Cognito, API Gateway